NexusIQ Corporation — Supply Chain Risk Assessment

FY 2025 Planning Cycle | Prepared by: Supply Chain Strategy | Date: February 10, 2025

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1. Purpose and Scope

This risk assessment evaluates supply chain vulnerabilities across NexusIQ's five product categories entering FY 2025. It incorporates lessons from the Q4 2024 Electronics inventory shortage documented in the Seasonal Demand Incident Report and the Inventory Shortage Root Cause Analysis, and provides a forward-looking view of supply risks that could impair NexusIQ's ability to fulfill demand across its 100-store network.

NexusIQ's supply chain serves $175,595,178.16 in annual revenue across 100,000 transactions. Electronics — the highest-revenue category at $90,958,302.29 — carries the highest supply chain concentration risk. The network's 72 below-reorder SKUs at year-end 2024 represent the most visible symptom of supply chain stress entering FY 2025.

2. Risk Inventory by Category

Category	Primary Risk	Severity	Likelihood	Risk Score	Key Vendor
Electronics	Single-vendor dependency (TechSource Global); Q4 demand volatility	HIGH	HIGH	9/10	TechSource Global
Home	Seasonal demand spikes; packaging quality issues	MEDIUM	MEDIUM	5/10	HomeSupply Co
Sports	Weather-dependent demand volatility	MEDIUM	LOW	3/10	SportPro Dist.
Clothing	Fashion cycle risk; high return rates driving reverse logistics cost	MEDIUM	MEDIUM	5/10	FashionLink
Food	Shelf life and cold chain compliance	LOW	LOW	2/10	FreshChain Ltd

3. Electronics Vendor Concentration Risk — Critical

The most significant supply chain risk facing NexusIQ in FY 2025 is the concentration of Electronics sourcing in TechSource Global. TechSource currently supplies approximately 85% of NexusIQ's Laptop, Tablet, and accessory inventory. The remaining 15% comes from spot purchases across three secondary vendors, none of which has the capacity to scale to meaningful NexusIQ volumes on short notice.

TechSource's FY 2024 on-time delivery rate of 94.2% — below the 95% contract minimum — combined with the Q4 2024 emergency order failure (two shipments arriving 7-8 days late during the critical Black Friday window) demonstrates that single-vendor dependency creates material business risk. If TechSource experiences a production disruption, NexusIQ has no qualified backup capable of filling the gap within a holiday-season lead time.

The corrective action — qualifying a secondary vendor at 15% contracted share — is on track for March 2025 completion. The target secondary vendor is Pacific Components Ltd, which completed initial quality assessment in January 2025 with acceptable results. A trial order of 500 Laptop units is scheduled for February 2025. If the trial is successful, the formal contract will target 15% Electronics share by Q2 2025, rising to 25% by Q4 2025 in advance of the next holiday season.

4. Clothing Category Reverse Logistics Risk

Clothing's 5,685-return contribution (assuming proportional share of the network total) and elevated return rates in Jeans (331 returns) and Jacket (287 returns) create a structural reverse logistics cost that erodes category margins. The FashionLink vendor is on performance notice for fill rate below target, adding forward logistics risk to an already stressed category.

Two risk mitigation options are under evaluation for Clothing in FY 2025. Option A is a return reduction program: investing in improved size guides, augmented reality try-on tools for the mobile app, and more accurate product photography to reduce returns driven by size and appearance mismatches. Estimated return rate reduction: 1.5-2.0 percentage points. Option B is a vendor change: replacing FashionLink with a vendor capable of higher fill rates and better product quality consistency. The decision will be made by the Q1 2025 business review.

5. Demand Forecast Risk for FY 2025

The revised forecasting methodology — using a 6-month rolling average weighted 60/40 toward recent periods — will reduce forecast error for high-growth categories. However, FY 2025 introduces a new forecasting challenge: NexusIQ's customer base has grown 11.8% in FY 2024 and the geographic expansion of the customer base into underserved North region markets is planned for H1 2025. New market demand is inherently harder to forecast than demand from established customer cohorts.

Supply chain planners should build 15% demand uplift buffers into North region inventory positions for the first two quarters of FY 2025 until demand patterns stabilize. This is consistent with the Inventory Reorder SOP's framework for new market inventory positioning. Total incremental inventory investment for North region expansion is estimated at approximately $840,000 in working capital above standard network norms.

6. Mitigation Roadmap — FY 2025

Initiative	Category	Target Completion	Expected Benefit	Investment
Secondary vendor qualification (Pacific Components)	Electronics	Q1 2025	Reduces single-vendor risk; adds surge capacity	$80K qualification cost
Forecast model update (6-month rolling, 60/40 weight)	All	Jan 2025 (complete)	Reduces Q4 demand surprise; earlier reorder triggers	Internal — no cost
Q4 seasonal multiplier in reorder SOP (1.3x)	Electronics, Home	Completed Mar 2024	Larger Q4 safety buffer	Carrying cost: ~$120K/year
Promotional demand elasticity review process	All	Feb 2025	Prevents bundle-driven stockouts like Q4 2024	Internal — 2 FTEs
FashionLink performance review / vendor change decision	Clothing	Q1 2025	Improved fill rate; reduced return-driven costs	TBD pending decision
North region expansion inventory buffer	All	Q1-Q2 2025	Supports new market launch without stockout risk	~$840K working capital

Progress updates to be presented at monthly Supply Chain Operations review. Major milestones reviewed at Q1 2025 business review.

7. Summary Risk Rating

Entering FY 2025, NexusIQ's supply chain is in a managed-risk state. The Q4 2024 shortage was a significant operational event but has produced a clear set of corrective actions with defined owners and timelines. The Electronics vendor concentration risk remains the highest-priority open item; until the secondary vendor relationship is established and proven, NexusIQ retains meaningful exposure to a repeat of Q4 2024 conditions in the next holiday season. The supply chain team considers the current risk trajectory as 'improving' provided the Q1 2025 vendor qualification milestones are met on schedule.